Drawing Lines in U.S. Monetary and Fiscal history

Thomas J. Sargent

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Abstract

This sequel to Sargent [2011] documents a catalogue of consequential reversals that illustrate the stability-versus-efficiency trade-off at the heart of that essay. That paper asked where to draw lines between (1) markets for money and credit, and (2) monetary and fiscal policies, showing that in both domains drawing sharp lines purchases price-level stability at a cost in economic efficiency, while erasing them promotes efficiency at the risk of instability. Milton Friedman moved from a 1948 proposal that implemented fiscal dominance over monetary policy to a 1960 proposal that implemented monetary policy dominance over fiscal policy; Volcker's 1979–82 attempt to implement a version of Friedman's 1960 policy lasted only a few quarters. Clay and Madison each opposed the First Bank on constitutional grounds, then became architects of the Second Bank once the War of 1812 exposed the fiscal cost of no national monetary institution. Salmon P. Chase engineered the Legal Tender Acts as Secretary of the Treasury, then as Chief Justice wrote the opinion in *Hepburn v. Griswold* (1870) declaring his own Acts unconstitutional. Keynes moved from defender of the gold exchange standard to author of its celebrated indictment; Eccles championed the wartime bond-price peg and then became the leading advocate of the Treasury-Fed Accord restoring monetary independence; Nixon closed the gold window and declared himself a Keynesian. Two institutional reversals complete the picture: Congress's half-century redrawing of the money–credit line from bond-backed National Bank Notes to the real-bills Federal Reserve Act; and Peel's Bank Charter Act of 1844, suspended in every major crisis within a generation of its passage. These reversals are evidence of how contextually sensitive, and how consequential, drawing and sustaining those lines can be.

1 Introduction

Sargent [2011] asked where to draw institutional lines between (1) markets for money and credit, and (2) monetary and fiscal policies. The essay argued that in both domains a genuine stability-versus-efficiency trade-off makes the choice difficult. Drawing a sharp line between money and credit—for example, requiring banks to hold 100-percent reserves or confining note issuance to government bonds only—can stabilize the price level, but creates rate-of-return wedges that are economically inefficient and invite regulatory evasion. The real bills doctrine, associated with Adam Smith, prescribes erasing that line by permitting banks to issue notes against sound short-term private claims; Sargent and Wallace [1982] showed in a general-equilibrium setting that such free intermediation can be Pareto superior to quantity-theory legal restrictions, while the same framework also reveals that the real bills criterion can be procyclical and destabilizing. On the monetary–fiscal side, the "unpleasant arithmetic" of Sargent and Wallace [1981] identifies an analogous tension: a monetary authority that commits credibly to a price-level path forecloses money creation as a fiscal instrument, purchasing stability at the cost of a constrained fiscal toolkit; a passive monetary authority accommodates fiscal needs but exposes the price level to fiscal pressures. In neither domain does has one of the polar arrangements been permanently sustained. Able American and British statesmen and economists i changed their minds, sometimes more than once. This essay examines five iAmericans' reversals in during the nineteenth century — Clay, Webster, Gallatin, Madison, Chase—and three in the twentieth century—Keynes, Eccles, Nixon—and then turns to two institutional reversals that unfolded over decades: Congress's step by step redrawing of the line between money and credit, from the bond-backed National Bank Notes of the 1860s to the real-bills-backed Federal Reserve notes of 1913; and, as a British parallel, Robert Peel's Bank Charter Act of 1844, whose statutory line was tested in three successive crises and bent without breaking.

Milton Friedman served as the primary example in Sargent [2011]. He moved from a 1948 proposal coordinating money creation with fiscal deficits to a 1960 fixed k-percent money-growth rule that insulates money entirely from the budget, trading seigniorage flexibility for price-level stability. The subsequent history of Friedman's rule adds a further chapter: Paul Volcker's Federal Reserve operationalized it in October 1979, only to abandon it by 1982 when financial deregulation destabilized

the very monetary aggregates on which the rule depended.

Henry Clay and James Madison present a complementary pattern at an earlier moment in American history. Both opposed the First Bank of the United States—Clay as a young Jeffersonian congressman on republican and states'-rights grounds, Madison as leading constitutional theorist of the founding generation. The War of 1812, fought without a national bank capable of mobilizing government credit, affected both of them. Two other conversions illuminate the same period: Daniel Webster, a New England Federalist who voted against the Second Bank's charter in 1816, became its most prominent legal and political defender within three years; and Albert Gallatin, the Jeffersonian firebrand who criticized Hamilton's Treasury from the House floor in the 1790s, became the Bank's strongest advocate when executive responsibility as Secretary of the Treasury convinced him of its fiscal indispensability. Clay would make a national bank a centerpiece of his "American System"; Madison would sign the bill creating the Second Bank in 1816, recognizing that long legislative practice had reframed his constitutional concerns from 25 years earlier.

Another individual reversal belongs to Salmon P. Chase. A hard-money Jacksonian Democrat who had opposed paper currency and any mixing of public and private credit, Chase became Abraham Lincoln's Secretary of the Treasury in 1861 and proceeded to preside over passage of the Legal Tender Acts of 1862 and 1863, which made inconvertible paper notes ($450 million in "greenbacks") legal tender for all debts. Appointed Chief Justice in 1864, Chase spent much of the rest of his career trying to undo his own handiwork. In *Hepburn v. Griswold* (1870) he wrote the majority opinion holding the Legal Tender Acts unconstitutional as applied to debts contracted before their passage. Grant's appointment of two new justices reversed that decision in *Knox v. Lee* (1871). Chase dissented.

The twentieth-century examples extend the pattern in instructive directions. John Maynard Keynes moved from sophisticated defender of the gold exchange standard to an outspoken opponent, driven by his reading of the deflationary costs that gold imposed on post-war Britain. His trajectory is a mirror image of Friedman's: both men were updating the same trade-off in response to the same historical events, but each concluded that the other's preferred failure mode was the greater danger. Marriner Eccles championed the wartime bond-price peg that made the Federal Reserve a fiscal instrument of the Treasury, then reversed course and became the leading institu-

tional advocate for monetary independence—at the cost of his chairmanship. Richard Nixon, campaigning on fiscal conservatism and sound money, closed the gold window and embraced Keynesian stimulus in a single Sunday announcement in August 1971.

Each reversal came with explicit accounts of why changed circumstances warranted a new institutional line. What the reversals collectively demonstrate is that the stability-versus-efficiency trade-off identified in Sargent [2011] involves real live political and constitutional choices that intelligent people have made differently in different contexts, and sometimes differently over the course of a single lifetime.

A coincident pattern deserves notice from the outset. The reversals documented here cluster around moments of systemic stress. The War of 1812 converted Clay and Madison simultaneously and vindicated Gallatin's earlier administrative conclusions. The Civil War broke Chase and funded the bond-backed currency system. The Great War and its deflationary aftermath drove Keynes's evolution. World War II created the circumstances that confronted Eccles. Nixon's 1971 reversal was in response to fiscal consequences of Vietnam War spending. This clustering suggests that the "forcing function" is not gradual evidence-accumulation alone, but crisis-induced revelations: costs of the existing institutional arrangement that were tolerable in a steady fiscal environment suddenly become undeniable to multiple actors.[1]

2 The Framework Recalled

The stability-versus-efficiency trade-off that organizes Sargent [2011] operates on both of the lines introduced above. On the money–credit side, the primary analytical tension is between quantity-theory narrow banking and the real bills doctrine. Drawing sharp lines—via 100-percent reserve requirements or bond-backed note issuance—stabilizes the price level but creates inefficient rate-of-return wedges and incentives for evasion. Sargent and Wallace [1982] showed that unfettered real-bills intermediation can be Pareto superior in general equilibrium, but also that the real bills criterion accommodates every credit expansion valued at current prices and so tends to amplify rather than dampen fluctuations. On the money–credit line, neither polar arrangement dominates.

[1] "Fiscal theories of price levels" were in the backs of the minds of the antagonists in our story. See Cochrane [2023] for a state-of-the-art presentation and critical analysis of such theories.

On the monetary–fiscal side, a useful accounting device is the government's flow budget constraint,

$$b_t + \tau_t + \mu_t m_{t-1} = g_t + (1 + r_{t-1})\, b_{t-1}, \tag{1}$$

where b_t is real debt, τ_t real tax revenues, g_t real spending, r_{t-1} the real interest rate on debt, and $\mu_t = (M_t/M_{t-1} - 1)$ the money-growth rate applied to a real money base m_{t-1}. Seigniorage $\mu_t m_{t-1}$ enters as a revenue source alongside taxes and borrowing, making the money supply and the price level potentially dependent on fiscal outcomes.

Two polar institutional arrangements can be distinguished:

Monetary dominance. The monetary authority commits credibly to a path for $\{M_t\}$ (or equivalently the price level), and the fiscal authority must choose $\{\tau_t,\, b_t,\, g_t\}$ to satisfy (1) without altering that path. Fiscal policy is the "residual claimant"; seigniorage is fixed exogenously by the monetary rule. Price-level stability is guaranteed by the commitment, but the fiscal authority cannot draw on money creation as a flexible revenue instrument.

Fiscal dominance. The fiscal authority sets $\{\tau_t,\, g_t,\, b_t\}$ autonomously and the monetary authority passively supplies whatever seigniorage is required to clear (1). The fiscal authority gains the flexibility to use money creation as a revenue instrument alongside taxes and borrowing. But fiscal dominance makes the price level a fiscal consequence; a deterioration in fiscal discipline is monetized.

As Sargent and Wallace [1981] showed with their "unpleasant monetarist arithmetic," neither arrangement dominates the other. Monetary dominance trades the option of seigniorage revenue for price-level stability; fiscal dominance trades stability for fiscal flexibility. Which side of the line is preferable depends on the fiscal environment, the credibility of institutional commitments, and the character of the shocks the government faces. The historical reversals documented below are illustrations.

3 Milton Friedman: From Fiscal Coordination to Monetary Rules

Friedman's 1948 paper in the *American Economic Review*, "A Monetary and Fiscal Framework for Economic Stability" [Friedman, 1948], proposed an institutional archi-

tecture in which monetary policy was explicitly subordinated to fiscal outcomes. The proposal rested on three pillars: (1) a requirement that commercial banks hold 100 percent reserves against deposits, so that money creation was entirely a government function; (2) automatic fiscal stabilizers (progressive taxation and transfer payments) that would cause the budget to move into surplus during booms and deficit during recessions without discretionary action; and (3) a rule specifying that any deficit be financed exclusively by issuing money, while any surplus be used to retire money. Under this system, the supply of money was mechanically linked to the fiscal balance: the government literally printed money to finance shortfalls and destroyed money when revenues exceeded expenditures.

This is a case of programmatic fiscal dominance. The monetary instrument—the money supply—is not controlled by an independent central bank pursuing a price-level target; it is set residually to satisfy a fiscal accounting identity. In the terminology of Sargent [2011], the monetary authority has been dissolved into the fiscal authority. Friedman embraced this design in 1948 because he believed that the economic cycle was predominantly driven by changes in aggregate demand, that discretionary monetary and fiscal management were dangerously unreliable, and that the automatic, formula-driven coupling of money creation to fiscal outcomes would deliver macroeconomic stabilization without giving policymakers enough discretion to do harm. His touchstone at this stage was the *coordination* of the two policy arms, not their separation.

By the late 1950s Friedman had moved to an opposite point of view. His Fordham University lectures, published as *A Program for Monetary Stability* [Friedman, 1960], advocated for a fixed k-percent annual growth rate in a well-defined monetary aggregate, entirely independent of fiscal conditions. The Federal Reserve should expand the money supply at a constant rate—Friedman suggested roughly 3 to 5 percent per year—regardless of whether the budget was in surplus or deficit. This is a textbook case of monetary dominance: the monetary rule is the binding commitment, and the fiscal authority must accommodate it.

Superficially, the distance between the two positions is large. Under the 1948 framework, a persistent fiscal deficit would automatically generate money creation and hence inflation; the 1960 framework explicitly blocks this channel. What changed?

Friedman identified several reasons [Friedman, 1960], Chapter 4. First, he had become convinced by the historical evidence, which he and Friedman and Schwartz [1963] were then assembling, that autonomous monetary shocks—changes in the money stock not driven by fiscal conditions—were the dominant source of macroeconomic fluctuations. The Great Depression was not a collapse of aggregate demand correctable by fiscal-monetary coordination; it was a monetary contraction engineered by Federal Reserve mistakes. If monetary shocks were the main danger, the right institutional response was to bind the monetary authority, not to couple it to fiscal outcomes.

Second, Friedman had concluded that fiscal discipline in the US polity was not sustainable. Governments that could finance deficits by money creation would reliably do so. The 1948 design assumed that the fiscal stabilizers would be genuinely automatic and symmetric: surpluses in good times would destroy money just as deficits in bad times created it. But the historical record suggested that democracies enlarged their fiscal commitments during downturns and failed to retire them during expansions. A design that coupled monetary policy to fiscal outcomes would therefore degenerate into a one-way ratchet toward inflation.

Third, and most interesting from the perspective of Sargent [2011], Friedman had reassessed the stability-efficiency trade-off. The 1948 framework prized efficiency (smooth automatic stabilization) and accepted the price-level risk entailed by fiscal coupling. The 1960 program prized stability (a committed monetary rule) and accepted the efficiency loss from foreclosing flexible seigniorage. The change was not a logical correction of an earlier error; it was a shift in the relative weight placed on the two objectives, driven by an updated reading of what the American political economy could actually sustain.

Friedman's reversal is a sharp illustration of the paper's main thesis: the optimal institutional line between fiscal and monetary authority depends on empirical and institutional features of the environment that are themselves changing and contestable.

3.1 Paul Volcker's experiment with Friedman's Rule

Friedman's k-percent rule was operationalized—at least temporarily—at the Federal Reserve. On October 6, 1979, Paul Volcker announced what the Fed called the "New Operating Procedures": rather than targeting the federal funds rate, the Federal

Open Market Committee would henceforth target non-borrowed reserves, in practice allowing interest rates to fluctuate freely while controlling the growth of monetary aggregates. The announcement was widely interpreted, including by Friedman himself, as the closest the American central bank had ever come to implementing the k-percent rule. Volcker had not adopted Friedman's specific prescription—he was targeting the path of money growth, not a fixed percentage—but the animating principle of the 1979 regime was identical to the 1960 proposal: bind the monetary instrument, accept short-term interest-rate volatility, and let the fiscal authority accommodate the monetary path rather than the reverse.

The monetarist experiment lasted until the summer of 1982—roughly thirty months. What ended it was not a failure of will on Volcker's part but an institutional problem that Friedman had identified in the abstract yet had not fully anticipated in practice. The financial deregulation of the early 1980s—the Depository Institutions Deregulation and Monetary Control Act of 1980, the proliferation of interest-bearing checking accounts (NOW accounts), and the growth of money-market mutual funds—fundamentally altered the demand function for the monetary aggregates that served as the rule's target. Velocity became unpredictable; the historical relationship between M1 growth and nominal income that had justified M1 as an intermediate target broke down. The aggregates were sending contradictory signals about the stance of policy.

By 1982, the recession induced by the disinflation was pushing unemployment toward 11 percent, the highest since the Great Depression. The collapse of Penn Square Bank in July 1982 and Mexico's announcement in August 1982 that it could not service its sovereign debt threatened systemic financial instability. Volcker, without formal announcement, shifted back to interest-rate targeting.

His memoir [Volcker and Harper, 2018] reflects explicitly on the episode. Volcker acknowledged that the monetary aggregates had become unreliable due to financial innovation, but also maintained that the 1979–82 disinflation had accomplished its primary objective—breaking inflationary expectations—even at the cost of an acute recession. In the terms of Sargent [2011], the Volcker experiment confirms the same lesson that Friedman's own reversal had taught: the optimal institutional design depends on features of the financial environment that change—sometimes faster than the institutions designed to manage them. Friedman had argued for the k-percent rule because he believed the money-demand function was stable enough to make the

monetary aggregate a reliable intermediate target. The deregulation of the 1980s undermined that stability, validating Friedman's rule in principle while rendering it unworkable in the specific institutional environment in which its single attempt at implementation occurred.[2]

4 Henry Clay: From Republican Critic to Champion of the American System

Henry Clay entered Congress as a Kentucky senator in 1806, steeped in the Jeffersonian Republican tradition that regarded the First Bank of the United States as an unconstitutional concentration of financial power in private hands. When the First Bank's twenty-year charter expired in 1811 and recharter was debated, Clay—now a newly elected member of the House—was among those who voted against it. The recharter bill died narrowly: the Senate divided 17–17, and Vice President George Clinton cast the deciding vote against; the House defeated it 65 to 64. Clay's opposition was rooted in the Jeffersonian political economy that Andrew Jackson would later systematize: the Bank was constitutionally suspect, an engine of privilege, and dangerous to the republic.

Five years later, Clay was among the foremost advocates in Congress for creating the Second Bank of the United States. The Second Bank was chartered in April 1816 with a twenty-year term, capitalized at $35 million (more than three times the First Bank's capitalization), and endowed with explicit authority to issue notes that could serve as currency nationwide. Clay incorporated the national bank into his "American System"—protective tariffs, federally financed internal improvements, and a national monetary institution—as necessary for American development and security.

What happened between 1811 and 1816 was the War of 1812. In the absence of a national bank, the Madison administration found it nearly impossible to finance the war. State banks proliferated and issued paper with no common standard; the government was reduced to borrowing at punishing terms from private financiers; military operations were hampered by the Treasury's inability to transfer funds efficiently across the continent. In 1814, state banks outside New England suspended

[2] Silber [2012] provides a detailed account of the 1979–82 episode and its bearing on the contest between monetary and fiscal authority. See p. **??** [page to be inserted].

specie payments—an event that dramatized the instability of a purely decentralized monetary system. The government's own fiscal operations were threatened: Treasury notes circulated at steep discounts, and the Secretary of the Treasury reported that he could not reliably meet the government's payroll.

Clay's reversal is intelligible as a response to this evidence. Before the War of 1812, the costs of a national monetary institution—the constitutional objections, the concentration of financial privilege, the threat to state banking interests—loomed large relative to its benefits, which seemed speculative and ideological. After the war, the corresponding cost-benefit calculation ran the other way: the demonstrated fiscal cost of the absence of a national institution was concrete and severe, while the constitutional objections had been blunted by the practical success of the First Bank during its twenty years of operation. The line between efficient national finance and the dangers of centralized monetary power had moved, not because the underlying principle had changed, but because the evidence on the costs of each institutional arrangement had accumulated.

In the framework of Sargent [2011], the First Bank had been an instrument of fiscal-monetary coordination: it held government deposits, facilitated tax collection, managed government debt, and provided a measure of monetary discipline through its willingness to redeem its notes. The two decades after its expiration amounted to an unplanned experiment in the absence of such coordination. Clay read the experimental results and updated accordingly. That is not inconsistency; that is rationality.

The sequel to Clay's conversion is worth noting. When Andrew Jackson vetoed the recharter of the Second Bank in 1832 and destroyed it by withdrawing government deposits in 1833–34, Clay emerged as the leading opponent of Jackson's action. The Bank War became the defining issue of Clay's Whig Party. The man who had voted against the First Bank in 1811 became, by the 1830s, the foremost defender of the principle of a national monetary institution against Jacksonian hard-money democracy. The reversal had become the fixed point from which he did not thereafter move.

4.1 Daniel Webster: From Opposing the Charter to Championing the Bank

Daniel Webster reversed from the opposite end of the partisan spectrum. Webster entered the House of Representatives in 1813 as a Federalist from New Hampshire—the party of Hamilton that had created the First Bank. Yet when the Second Bank bill came before the House in 1816, Webster voted against it. His objections were not Clay's earlier Jeffersonian ones about constitutionality or the concentration of privilege; Webster opposed the *specific terms* of the bill, arguing that it failed to require prompt resumption of specie payments and that the Bank's capital structure and relationship to the government were poorly designed. He wanted a sounder national bank, not no national bank at all.

Within three years Webster had become the Bank's foremost legal champion. He argued the Bank's case before the Supreme Court in *McCulloch v. Maryland* (1819), the decision that vindicated the constitutionality of a national bank under the Necessary and Proper Clause and became a cornerstone of federal implied-powers doctrine. By the 1820s, Webster was on standing retainer as the Bank's counsel, a relationship he maintained throughout Nicholas Biddle's presidency of the institution. The financial entanglement was not incidental: Webster borrowed from the Bank and was frank about the connection, writing to Biddle on at least one occasion to note that "my retainer has not been renewed or refreshed as usual" and to inquire whether the Bank wished the relationship to continue. During the Bank War of the 1830s, Webster stood beside Clay as one of the two principal Senate opponents of Jackson's campaign to destroy the institution.

Webster's reversal is less ideological than Clay's—he moved from objecting to a specific design to championing the institution, rather than crossing a philosophical divide—but illustrates an additional mechanism of conversion. Where Clay was compelled by the evidence of fiscal catastrophe during the War of 1812, Webster was converted by the Bank's *operational success* in its early years and by his growing appreciation, as the leading commercial lawyer of New England, of what a well-managed national monetary institution could accomplish for interstate commerce and the stability of private credit. The institution that looked ill-designed on paper in 1816 proved indispensable in practice by 1819. In the framework of Sargent [2011], Web-

ster updated his assessment not of the *principle* of fiscal-monetary coordination but of the *institutional capacity* to execute it competently—precisely the kind of empirical updating that makes the optimal location of the line a moving target.

5 James Madison: Constitutional Principle and the Doctrine of Liquidation

James Madison's trajectory runs parallel to Clay's and deeper: his opposition to the First Bank was constitutional, grounded in the theory of his generation.

When Alexander Hamilton proposed the First Bank in January 1791, Madison led the House opposition. His speech of February 2, 1791 is one of the classic statements of strict constructionism. The Constitution granted Congress enumerated powers. The power to charter a corporation was not among them. The "necessary and proper" clause could not be interpreted to extend congressional power to any means that might be deemed convenient or useful; "necessary" had to mean genuinely indispensable to the execution of an enumerated power, and chartering a bank was not indispensable to anything the Constitution explicitly authorized. Madison also invoked what would later be called the principle of original understanding: the framers' intentions, as manifested in the ratification debates, gave no support for so broad an implied power.

Hamilton's counter-argument won: Washington accepted Hamilton's opinion and signed the bill. Madison's position established the intellectual framework that Jeffersonian Republicans would invoke for the next two decades, and that Jackson's Democrats would recycle in the Bank War of the 1830s.

5.1 Albert Gallatin: From Jeffersonian Critic to Executive-Branch Champion

Albert Gallatin's trajectory illustrates the same conversion from the vantage point of executive responsibility. Gallatin arrived in Congress in 1795 as a Jeffersonian Republican from western Pennsylvania, steeped in the agrarian suspicion of Hamilton's financial architecture. As the leading Republican voice on fiscal affairs in the House, Gallatin subjected Hamilton's Treasury operations—including the operations of the Bank—to close scrutiny, viewing them as instruments of Federalist political consoli-

dation rather than as neutral fiscal machinery. His opposition was pragmatic rather than constitutional: where Madison had argued that chartering a bank exceeded Congress's enumerated powers, Gallatin objected that Hamilton's Bank served Federalist patronage networks more than it served the public interest.

When Jefferson appointed Gallatin Secretary of the Treasury in 1801, Gallatin brought these Jeffersonian convictions to the office that Hamilton had designed. Administrative experience changed his mind rapidly. The Bank proved to be not the engine of partisan privilege that Gallatin had attacked from the House floor but an indispensable instrument of day-to-day fiscal management: it held government deposits safely, transferred funds efficiently across the continent, facilitated customs collections at dispersed ports, and provided a stable medium for servicing the national debt. By the time the Bank's twenty-year charter approached expiration, Gallatin had become its strongest advocate within the administration. In a formal report to Congress in 1809, he urged recharter in terms that would have been unthinkable from the Jeffersonian firebrand of the 1790s, arguing that the Bank was "necessary" for the responsible management of the government's finances and warning that its dissolution would impose severe fiscal costs.

The warning went unheeded. The recharter bill died in 1811—by the narrowest of margins, as noted in the preceding section—and the fiscal catastrophe of the War of 1812 vindicated Gallatin's prediction in every particular. Gallatin, who served as a peace commissioner at Ghent in 1814, watched from abroad as the Treasury struggled with exactly the problems he had foreseen: no reliable depository for government funds, no mechanism for interstate fiscal transfers, no institutional anchor for public credit. The experience confirmed what his years at the Treasury had already taught him: that the abstract Jeffersonian case against a national bank could not survive contact with the operational realities of governing a continental republic. Gallatin's later writings on banking and currency, notably his *Considerations on the Currency and Banking System of the United States* (1831), drew explicitly on his executive experience to defend the principle of a national monetary institution against both Jacksonian hard-money populism and the proliferating disorder of unregulated state banking.

Gallatin's conversion was driven not by wartime emergency—which came later—but by the accumulation of administrative evidence. The War of 1812 confirmed what

Gallatin already knew; it did not cause his reversal. In the language of Sargent [2011], Gallatin's experience as Treasury Secretary updated his estimate of the *efficiency gains* from fiscal-monetary coordination: the institution that had appeared, from the opposition benches, to be a source of corruption and privilege turned out, from the Treasury desk, to be a source of fiscal efficiency that no combination of state banks and ad hoc arrangements could replicate.

5.2 Madison's Reversal

In 1815, near the end of the war and his presidency, Madison vetoed a bank bill—not on constitutional grounds, but because the proposed bank lacked features that would make it an effective fiscal instrument for the government. The veto message explicitly declined to reopen the constitutional question, which Madison said had been settled by congressional and executive practice over twenty-four years. The following year, in 1816, Madison signed the bill creating the Second Bank of the United States.

How did the author of the 1791 speech against bank constitutionality come to sign the bill creating a larger successor? Madison's answer was sophisticated and, to his critics, convenient. He had articulated it already in the 1815 veto message: while his own constitutional views had not changed, the Constitution's meaning was not fixed solely by the original understanding of individual framers. It was also determined by the accumulated legislative, executive, and judicial practice of the republic. A course of action repeatedly approved by Congress, acquiesced in by successive presidents, and never invalidated by the courts thereby acquired constitutional legitimacy through what Madison called the process of "liquidation"—the settlement of constitutional meaning by institutional practice over time.[3] By 1816, twenty-four years of the First Bank's operation had, in Madison's view, liquidated the constitutional question in favor of congressional authority.

The liquidation doctrine is consequential. It implies that the institutional line between

[3]Madison used the term "liquidation" for this process as early as *Federalist* No. 37 (1788), where he wrote that the meaning of constitutional terms would be "liquidated and ascertained by a series of particular discussions and adjudications." He returned to it in correspondence of the 1820s–30s to justify his acceptance of the Bank's constitutionality through long legislative practice. For modern scholarly treatments see Nelson [2003] and Baude and Sachs [2017]. An economist will notice a double entendre: "liquidation" is also what banks do—converting illiquid assets into liquid (easily tradable) liabilities. Madison's constitutional usage and the banker's usage share the same root metaphor: something fixed and opaque becomes fluid and transparent through use.

permissible and impermissible concentration of fiscal-monetary power is not fixed by original text alone; it evolves through a common-law-like process of precedent and practice. Institutions that were once constitutionally contested can become constitutionally settled, and the initial objections correspondingly lose their force. In the terms of Sargent [2011], the boundary between stability-promoting and efficiency-promoting arrangements is not only analytically ambiguous; it is constitutionally labile, subject to reinterpretation as practice accumulates.

Madison's reversal reflects the same lesson Clay drew from 1812: the War had revealed the fiscal consequences of the absence of a national monetary institution. A republic that could not finance its wars without submitting to ruinous private financiers was ill served by constitutional principles, however impeccably derived.

The sharpest contrast to Madison is Jefferson, who never accepted the liquidation doctrine and insisted to the end of his life that the Bank was unconstitutional. Jefferson's consistency and Madison's flexibility represent the two sides of a live debate: whether institutional practice can revise constitutional meaning, or whether original principle must constrain practice however inconvenient the results.

6 Salmon P. Chase: From Hard Money to Legal Tender to Constitutional Revision

The most dramatic reversal belongs to Salmon P. Chase. His career traces a complete arc: from principled hard-money doctrine, through wartime necessity, to a judicial crusade to undo his own executive work. The fullest contemporary anatomy is George Bancroft's *A Plea for the Constitution of the United States Wounded in the House of Its Guardians* [Bancroft, 1886].

6.1 A Witness of Singular Authority

Bancroft (1800–1891) brought to monetary constitutionalism not the moralism of a hard-money partisan but the documentary authority of a man who had spent fifty years in the primary sources of the founding era. He had personally known Madison and Adams; had served as Secretary of the Navy under Polk and as Minister to Prussia and Germany under Johnson and Grant; and was 84 when, in the wake of

Juilliard v. Greenman (1884), he wrote his *Plea*. The book is a historical argument that the framers at Philadelphia had deliberately and by large majorities stripped from the federal government any power to make paper money legal tender for private debts—and that a sequence of Supreme Court decisions had betrayed that design. His central exhibit, running through every part of the work, is Salmon P. Chase.

Bancroft's constitutional argument rests on what happened in Philadelphia on August 16, 1787. The original draft of the Constitution had given Congress the power "to borrow money, and emit bills on the credit of the United States." Gouverneur Morris moved to strike the words "and emit bills." The motion passed 9–2. Madison's convention notes recorded that the majority "thought it advisable not to insert this power; thinking it probable that the omission would not disable the use of public notes so far as they would be safe and proper, and would only cut off the pretext for a paper currency, and particularly for making the bills a tender either for public or private debts." Article I, Section 10 then explicitly forbade the states from emitting bills of credit or making anything but gold and silver coin legal tender. For Bancroft, this two-sided design—the power denied to Congress by omission, the power denied to the states by express prohibition—was the clearest possible evidence that the framers had closed the door to paper legal tender at every level of government.

Chase's career is, in Bancroft's telling, the story of that door being pried open and the constitution bleeding as a result.

6.2 The Hard-Money Democrat

Chase began his public career in the 1830s embedded in the Jacksonian hard-money tradition: paper money not convertible to specie was fraud; bank-note issuance was an illegitimate privilege; sound money meant gold and silver coin, or notes immediately redeemable in them. Chase was a Free Soiler and then Republican by the 1850s, but his monetary convictions remained Jacksonian: he distrusted paper, trusted specie, and argued that mixing governmental and banking functions corrupted both.

As Governor of Ohio (1856–60) and then Senator, Chase never departed from these positions. When Lincoln appointed him Secretary of the Treasury in March 1861, Chase brought this monetary philosophy with him. His first instinct was to finance the Civil War through taxation and borrowing at market terms, without resort to

inconvertible paper. He raised tariffs, promoted the new income tax, and placed bond issues with Jay Cooke and other financial houses. In his initial communications to Congress, Chase explicitly rejected proposals for a legal-tender paper currency.

6.3 The Legal Tender Acts and the 5-20 Bonds

By December 1861 the situation had become critical. Banks had suspended specie payments; the Treasury could not meet its obligations; military expenditures exceeded what bond markets could absorb at sustainable rates. Chase reversed course. He endorsed, designed, and lobbied for the Legal Tender Act of February 25, 1862, which authorized the issuance of $150 million in Treasury notes—"United States Notes," quickly styled "greenbacks"—that were legal tender for all debts, public and private, but not convertible to specie. Subsequent acts in July 1862 and March 1863 authorized additional issues, bringing the total to approximately $450 million.

Alongside the greenbacks, Chase issued the so-called 5-20 bonds to finance the war: six-percent coupon bonds redeemable after five and payable within twenty years, whose authorizing statute (the Act of February 25, 1862) specified that interest would be paid in coin but was entirely silent on principal. Chase bridged this statutory gap with reputational commitment, marketing the bonds to the public—through Jay Cooke's retail campaign—with the implicit understanding that principal, too, would eventually be repaid in gold. The commitment was extra-statutory: it rested on Chase's personal credibility as a hard-money man, not on any legislative guarantee. That would matter later.

Chase's arguments for the Acts were purely fiscal and pragmatic: the government faced a choice between legal-tender paper and either national bankruptcy or negotiated independence. He invoked the Revolutionary War precedent and argued the power to issue legal-tender notes was inherent in sovereignty. He did not pretend consistency with his earlier views; he argued only that emergency had foreclosed the options he would have preferred.

For Bancroft, the Revolutionary precedent was precisely the wrong one: continental dollars were the reductio ad absurdum of paper money and the primary inspiration for the Philadelphia framers' decision to close the door. To cite them as precedent was to cite the disease as justification for a recurrence.

In the taxonomy of Sargent [2011], the Legal Tender Acts were a decisive move toward fiscal dominance. The government issued an inconvertible currency and declared it legal tender; the "monetary policy" embedded in this decision—the price level would be whatever was required to make $450 million in greenbacks cover real government expenditures—was entirely subordinate to fiscal necessity. Chase had, under pressure of circumstances, crossed from the hard-money extreme of monetary dominance (specie standard, no fiat paper) to the fiscal-dominance extreme (inconvertible legal-tender paper issued at the government's discretion).

6.4 *Hepburn v. Griswold* (1870): Chase's First Reversal

Lincoln appointed Chase Chief Justice in December 1864. For the remainder of his life, Chase pursued a constitutional agenda centered on reversing the monetary consequences of his own Treasury policy.

The occasion came in *Hepburn v. Griswold* (75 U.S. 603), decided February 7, 1870. The case involved a debt of $11,250 contracted in 1860, before the Legal Tender Act, and tendered for payment in 1864 with greenbacks. The creditor, Mrs. Hepburn, argued that she was entitled to gold coin equivalent, since the contract antedated the Act that made paper legal tender. By a 4–3 vote (two seats were vacant), the Court held for the creditor. Chase wrote the majority opinion.

The *Hepburn* opinion rested on two grounds. First, Chase argued that compelling creditors to accept depreciated paper for pre-existing gold debts deprived them of property without due process in violation of the Fifth Amendment. Second, he argued that making greenbacks legal tender for such debts was not within the Necessary and Proper Clause: Congress could permissibly issue paper currency to finance the war, but the further step of compelling private creditors to accept it for pre-existing obligations was not "necessary" to any express power in the required sense. Crucially, Chase appealed not to express textual prohibition but to the "spirit" of the Constitution—a methodological choice that would shortly be turned against him.

The opinion conspicuously did not address whether Chase had been correct in 1862 to recommend the Acts. But the logical entailment was inescapable: if making greenbacks legal tender for pre-existing debts was unconstitutional, then Chase had committed a constitutional error as Secretary. Justice Samuel Miller's dissent noted that

the majority had struck down legislation that the Chief Justice himself had recommended and defended as a member of the executive branch. Miller also criticized Chase's appeal to the "spirit" of the Constitution as "abstract and intangible" reasoning that invited an unlimited judicial veto over congressional action.

6.5 *Knox v. Lee* (1871): The Court-Packing Reversal

The reversal of *Hepburn* came with stunning speed and in circumstances whose constitutional impropriety has been debated ever since. On the very day *Hepburn* was announced—February 7, 1870—President Grant nominated two new justices to fill existing vacancies: William Strong of Pennsylvania and Joseph Bradley of New Jersey. Chase believed, and said, that Grant had deliberately timed the nominations to reconstitute the Court and obtain a reversal. When the Court reached its new complement of nine, the Attorney General moved for reargument of the legal tender question. *Knox v. Lee* and *Parker v. Davis* (79 U.S. 457) were decided together on May 1, 1871, by a vote of 5–4, overruling *Hepburn*.

Justice Strong's majority opinion relied on the implied-powers doctrine of *McCulloch v. Maryland*: if Congress has the war power and the power to borrow money, and if a legal-tender paper currency is an appropriate means of executing those powers, the Acts are constitutional under the Necessary and Proper Clause. Strong's opinion included a passage chiding Chase for having "forced" an earlier decision when the Court was divided and on the verge of receiving new appointees— the majority lecturing the former majority about the indecorum of the process by which the current majority had come to exist.

Chase dissented, joined by Justices Clifford and Field. His dissent insisted that the majority had abandoned the constitutional limitations on congressional power in favor of an open-ended theory of implied powers that left no meaningful constraint on federal action. Field's dissent was constitutionally precise: the doctrine that whatever is not expressly forbidden may be exercised would "change the whole character of our government" from one of limited, enumerated powers to one of unlimited powers in its relations to the people.

Bancroft regarded the Court's composition question as corroborating his thesis. On a constitutional question of the first magnitude—whether the federal government may

compel private citizens to accept depreciated paper in discharge of gold obligations—the outcome turned not on any change in the law or the facts but on the personnel of the Court. This was, in constitutional terms, not a "hard constraint" (in the sense of a textual prohibition too clear to circumvent) but a "soft constraint" that yielded to political pressure once the political composition of the appointing authority changed. The same logic that had made Chase's reputational commitment on the 5-20 bond principal extra-statutory and therefore fragile had made the constitutional prohibition on legal tender extra-textual and therefore fragile.

6.6 *Juilliard v. Greenman* (1884): Legal Tender as Permanent Sovereignty

The third case swept away whatever remained of the *Hepburn* principle. In *Juilliard v. Greenman* (110 U.S. 421), decided March 3, 1884, Justice Gray's majority opinion—joined by eight of nine justices—held that the power to declare paper money legal tender was not limited to wartime emergencies but was a permanent attribute of national sovereignty. The statutory notes at issue had originally been issued during the war, then redeemed and reissued in peacetime under the Resumption and Refunding Act of 1878; no emergency justified them. That did not matter. Gray wrote:

> The power to make the notes of the government a legal tender in payment of private debts being one of the powers belonging to sovereignty in other civilized nations, and not expressly withheld from congress by the constitution; we are irresistibly impelled to the conclusion that the impressing upon the treasury notes of the United States the quality of being a legal tender in payment of private debts is an appropriate means, conducive and plainly adapted to the execution of the undoubted powers of congress.

For Bancroft, this language was "the language of revolution." The argument from what "other civilized nations" had done had no place in interpreting a constitution deliberately designed to withhold powers that every European monarchy exercised. The argument from "not expressly withheld" converted a government of enumerated powers into one of unlimited powers: if Congress may do anything not expressly forbidden, then the framers' enumeration of powers was a dead letter. Moreover,

the same nine justices who joined the *Juilliard* majority had unanimously declared, just two years earlier, that "the government of the United States is one of delegated, limited, and enumerated powers," and that "every valid act of congress must find in the constitution some warrant for its passage." Bancroft noted that the *Juilliard* opinion was therefore "in flagrant antagonism" to the Court's own recent, unanimous, and repeatedly given interpretations of the constitutional structure.

The sole dissenter was Justice Field—in Chase's *Hepburn* majority and in the *Knox* dissent, now standing entirely alone. Bancroft insisted Field was not a minority of one but the transmitted judgment of the founding generation: beside him stood "invincible vouchers for the rightness of his dissent"—Wilson, Ellsworth, and Paterson, all appointed to the Court by Washington from among the Convention's framers.

6.7 Constitutional Tragedy and Soft Constraints

Bancroft read Chase's trajectory as symptomatic of a constitutional culture that had lost its moorings, not as individual hypocrisy. But his framing makes the theoretical point with unusual clarity. Chase's reputational commitment to gold repayment of the 5-20 bond principal and his constitutional commitment to the invalidity of legal-tender paper were both *extra-statutory*—both existed as assurances given without explicit legislative or textual authority. Both were undone by the same political logic that had originally forced Chase to accept the Legal Tender Acts as a war measure. When the political environment changed enough to matter, neither commitment was self-enforcing.

For the purposes of Sargent [2011], the arc of Chase's career encapsulates the full range of the stability–efficiency trade-off. His early Jacksonian position was an extreme form of monetary dominance: specie only, no government paper, the fiscal authority must adjust entirely to monetary reality. The Legal Tender Acts represented a wartime capitulation to fiscal dominance: the monetary instrument becomes whatever fiscal survival requires. His judicial career was an attempt to restore monetary dominance by constitutional command—to fix in law the institutional line against which executive necessity, however acute, could not encroach.

The *Hepburn–Knox–Juilliard* trilogy demonstrates that the attempt failed, and *why* it failed. The constitutional constraint Chase sought to enforce was a soft constraint:

it rested not on language expressly prohibiting the legal tender power but on the "spirit" of the Constitution as Chase interpreted it, and on the historical evidence that the Philadelphia Convention had deliberately struck out the "emit bills" clause. Bancroft thought this soft constraint should have been treated as a hard one; the Court disagreed, and the political environment— unwilling to foreclose a fiscal instrument it had used to win a civil war—did not sustain the harder reading. Institutional lines that are not politically sustainable cannot be maintained by constitutional command alone, and Chase's career is the clearest proof.

7 The Money–Credit Line: From Bond-Backing to Real Bills, 1863–1913

The reversals catalogued above concern the boundary between monetary and fiscal *authorities*. A related contested line—between *money* and *credit*—was redrawn repeatedly by Congress in the half-century following Chase's National Bank Acts.

7.1 Bond-Backing and the Line at Government Debt

The National Bank Act of 1864 drew the money–credit line at an unusual location: United States government bonds. National Bank Notes—the dominant hand-to-hand currency of the post-Civil War era—could be issued only against specified federal bonds deposited with the Comptroller of the Currency. That institutional officer, whose primary function was verifying the bond portfolios, could certify the soundness of every dollar in circulation. The line between what counted as "money-eligible" and what counted as mere "credit" was the difference between government bonds on one side and every other asset—commercial loans, railroad securities, warehouse receipts, farm mortgages—on the other.

In the framework of Sargent [2011] this arrangement is a form of *monetary dominance*, but with a peculiar twist: the binding constraint on the money supply was not a target set by an independent central bank but the stock of outstanding federal debt. When the government ran the large primary surpluses of the 1870s and 1880s and retired war bonds, the supply of eligible collateral contracted, and the currency supply contracted with it regardless of any commercial demand for more money. Fiscal virtue, no less

than fiscal profligacy, could thus distort the monetary instrument; the two distorted it in opposite directions, but neither direction was chosen for monetary reasons. The bond-backing requirement was a hard constraint against inflationary over-issue, but it purchased that stability by tying the money supply to the government's debt-management calendar rather than to the needs of trade.

7.2 The Inelasticity Debate as Four Decades of Disagreement about the Line

Critics from the 1870s onward called the resulting currency "inelastic." The agrarian South and West needed sharply higher volumes of currency during planting and harvest seasons; the bond-backed system offered no mechanism for such expansion. Financial panics punctuated the era. Two coalitions proposed to move the money–credit line in opposite directions.

Greenbackers and silver inflationists wanted to dissolve the line almost entirely. If the government simply issued more fiat currency—greenbacks to be increased rather than retired, or silver to be coined at a generous ratio—there was no need for any specific asset to "back" the money at all. This was, at the limit, the same fiscal dominance that Chase had temporarily embraced during the war: the money supply would be whatever fiscal or political convenience required. William Jennings Bryan's "Cross of Gold" speech of 1896 crystallized this position in its most direct form.

Asset-currency reformers drew the line elsewhere. Their complaint was not too little money but the wrong kind of backing. They argued that national banks should be permitted to issue notes against a wider range of productive assets—railroad and corporate bonds, agricultural warehouse receipts, or any "good assets" on a bank's balance sheet—rather than against government bonds alone. This was a proposal to relocate the money–credit boundary from "government debt only" to "any sufficiently sound private claim." By broadening the eligibility set, they hoped, the currency supply would be free to expand with productive activity rather than with the Treasury's debt schedule.

The Aldrich–Vreeland Act of 1908, passed in the wake of the Panic of 1907, was a temporary halfway house. It allowed groups of national banks to issue emergency currency backed by "almost any securities the banks were holding," but imposed a

punitive tax schedule to ensure rapid retirement once conditions normalized. More consequentially, it created the National Monetary Commission, chaired by Senator Aldrich, whose multi-volume studies of European central banking provided the intellectual scaffolding for the Federal Reserve Act of 1913.

7.3 The Real Bills Resolution and Its Costs

The Federal Reserve Act resolved the inelasticity debate by drawing the money–credit line at a new location: short-term, self-liquidating commercial paper. Section 13 of the Act confined Federal Reserve discount operations to notes and bills "arising out of actual commercial transactions" maturing within ninety days, and explicitly excluded paper drawn to finance positions in "stocks, bonds, or other investment securities." The so-called *real bills doctrine*— also called the commercial loan theory of banking, with roots in Adam Smith's *Wealth of Nations* and a long subsequent history—became the operating principle of the new institution.

The new line distinguished money from credit not by asset class but by the *purpose and duration* of the underlying transaction. A sixty-day bill drawn against a shipment of grain was eligible because the transaction was real, productive, and self-liquidating: when the grain was sold, the bill would be repaid and the money supply would contract automatically. A call loan financing stock market speculation was ineligible. In the political rhetoric of the era the line ran between Main Street and Wall Street; in the theoretical rhetoric it ran between "money backed by real production" and "money backed by finance."

The real bills criterion solved the inelasticity problem: the money supply could expand with productive activity rather than with the federal debt stock. But it imported a different instability, one Henry Thornton had identified in 1802 in his *Paper Credit of Great Britain* [Thornton, 1802]: by accommodating every credit demand arising from commercial transactions valued at *current prices*, the criterion set up a positive feedback from prices to money creation to prices. In a boom, prices rise; the nominal value of commercial transactions rises; the demand for discount rises; and the Fed, faithfully applying the real bills criterion, expands the money supply—adding fuel to the expansion rather than restraining it. The criterion is inherently procyclical.

The consequential application came in 1929–1933. The Federal Reserve Board mem-

ber Adolph Miller launched a "Direct Pressure" policy: any bank seeking discount window relief had to certify that it had never made speculative loans. Banks, unwilling to submit to the implied interrogation, refrained from borrowing reserves even in the face of withering deposit outflows. Instead of discounting at the window, banks failed—nearly 9,000 of them. Friedman and Schwartz [1963] held the resulting one-third contraction of the money stock largely responsible for the Great Depression. Where the bond-backed currency of the national banking era had been passively contractionary in the face of fiscal surpluses, the real bills currency proved passively contractionary in the face of banking panics. Each line carried its own characteristic failure mode.

7.4 The Theoretical Afterlife and the Sargent–Wallace Reconsideration

The real bills debate connects directly to the framework of Sargent [2011]. In a 1982 article, Sargent and Wallace [1982] showed that within overlapping-generations models the real bills prescription—unfettered private intermediation against productive assets—can be Pareto superior to the quantity-theory alternative, which imposes restrictions on intermediation in order to control the price level. This is a version of the stability–efficiency trade-off that Sargent [2011] identifies in the fiscal–monetary domain: the quantity-theory regime purchases price-level stability at the cost of constrained intermediation; the real bills regime allows efficient intermediation at the cost of weaker price-level control. Laidler [1984] objected that the Sargent–Wallace version had abstracted away the key feature of the historical doctrine—the restriction to specific collateral types—but the theorem demonstrated that the money–credit distinction is not merely an institutional accident or a historical curiosity. It reflects a genuine analytical trade-off whose terms depend on the structure of the economy and the quality of the regulatory environment.

The congressional progression from bond-backed currency to real bills—from the National Bank Acts of 1863–64 to the Federal Reserve Act of 1913, through four decades of inelasticity complaint and asset-currency agitation—thus illustrates in miniature the same dynamic as the individual reversals catalogued in this paper. The actors who drafted the Federal Reserve Act were consciously reversing the institutional design that Chase had supervised as Secretary of the Treasury. Their reversal solved

one class of problems while creating another; and the real bills doctrine embedded in the new Act would itself require substantial revision in the Banking Acts of 1932–35, a revision precipitated by the very failure mode that Thornton had foreseen in 1802. The half-century arc from National Bank Notes to Federal Reserve notes is another turn of the same wheel: each new institutional line is drawn in response to the failures of the previous one, and each generates its own successor crisis.

7.5 Peel's Bank Charter Act and Its Three Suspensions: A British Parallel

The dynamic of statutory lines tested and bent by crisis has a precise British parallel in the Bank Charter Act of 1844. Robert Peel's Act divided the Bank of England into two departments: an Issue Department, which could issue notes only against gold or a fixed fiduciary limit of £14 million; and a Banking Department, which could lend freely but had no special link to note issuance. The Currency School theorists who designed this structure—Samuel Jones Loyd (Lord Overstone) was their principal theorist—intended it as a strict statutory constraint on money creation [Fetter, 1965]. By tying the note issue to gold and banning discretionary accommodation, they believed they had drawn the line against inflationary over-issue once and for all.

The Act was suspended by Treasury letter in October 1847, November 1857, and May 1866—in every major financial panic within a generation of its passage. In each case the mechanism was the same: the Bank of England, constrained by the Act from expanding its note issue, faced a choice between watching the banking system collapse and requesting from the government an indemnity letter authorizing temporary breach of the statutory limit. The government wrote. In 1847, the announcement alone proved sufficient to calm the panic; the Bank's note issue never in fact breached the legal maximum. In 1857, following the failure of the Western Bank of Scotland and the Glasgow City Bank, the Bank did exceed its fiduciary limit and Parliament subsequently provided indemnity. In 1866, after the collapse of Overend, Gurney and Company—then the largest bill broker in London—the Bank initially hesitated to lend freely, exacerbating the panic before the suspension letter arrived.

Walter Bagehot's response in *Lombard Street* [Bagehot, 1873] is the consequential commentary on this pattern. Bagehot was not an opponent of the Bank Charter Act

on Currency School grounds; he accepted its basic institutional structure as too politically entrenched to repeal. But his prescription inverted Peel's design. Where Peel's Currency School architects had intended the gold constraint to force contraction in a crisis—compelling banks to liquidate and prices to fall, rather than accommodating speculative excess—Bagehot argued that in a genuine panic the Bank *must* lend freely at a penalty rate against good collateral. The lender-of- last-resort function that *Lombard Street* articulated was the theoretical mirror image of everything the Currency School had built the Act to prevent.

In the terms of Sargent [2011], the Bank Charter Act was an attempt to inscribe monetary dominance into statute: the note issue would be bound, the banking or fiscal residual would adjust. The three suspensions demonstrated that the attempt produced a statutory line rather than an institutional reality. The line was not changed; it was, as Bagehot saw, managed by the publicly known expectation that it would be suspended whenever truly tested. "The practical consequence is that the Bank of England is obliged to keep the reserve for the whole banking system," Bagehot wrote, "and yet no one has said it should; and no one will say it should." The constraint existed on paper; the practice had already moved past it.

The parallel to the *Hepburn–Knox–Juilliard* trilogy is exact. In both cases a monetary rule was established by law or constitutional convention to prevent fiscal or speculative accommodation. In both cases the rule was suspended under pressure—by executive letter in Britain, by court-packing in America. In both cases the exception tended to become the norm: repeated Bank Act suspensions made suspension a publicly anticipated contingency, just as *Juilliard* made legal tender a permanent attribute of sovereignty. And in both cases the analytical response was not a defense of the original line but a new doctrine that institutionalized the exception: Bagehot created the lender-of-last-resort doctrine to rationalize the accommodation; Bancroft could only mourn that the Philadelphia framers' judgment had been overridden. The difference in outcome reflects a difference in institutional creativity: Britain produced a reformulation that preserved the spirit of stability while accepting the necessity of crisis lending; America consumed the original constraint without producing a comparably coherent successor.

8 Twentieth-Century Reversals

Three twentieth-century episodes extend the same pattern and illuminate dimensions of the stability–efficiency trade-off that the earlier record leaves implicit.

8.1 John Maynard Keynes: From Gold Standard Defender to Its Foremost Critic

Among economists, the sharpest reversal of the twentieth century belongs to Keynes. Its direction is the mirror image of Friedman's: where Friedman moved from fiscal-monetary coordination toward a strict monetary rule, Keynes moved from defense of a specie-based constraint toward managed money and fiscal flexibility.

Keynes's starting point was *Indian Currency and Finance* [Keynes, 1913]. The book analyzed India's gold exchange standard affirmatively: the system was a rational institutional evolution that captured the price-stability benefits of gold convertibility without requiring a full gold circulation. The book was received as evidence of a youthful attachment to specie discipline [Moggridge, 1992].

Within a decade the world that book described had been destroyed by the Great War. Britain had suspended gold convertibility in 1914; the United States had accumulated the bulk of the world's monetary gold; wartime inflation had restructured debt relationships in ways that made the restoration of pre-war parities a program of deflation rather than of price stability. By 1923, when Keynes published *A Tract on Monetary Reform* [Keynes, 1923], his position had inverted. The gold standard was "already a barbarous relic"; the objective was to manage the price level directly through central bank discretion rather than subordinate domestic conditions to an external specie constraint. The book proposed a "managed currency" targeting price-level and output stability. In the taxonomy of Sargent [2011], this is a shift from monetary dominance toward managed discretion—the monetary authority responding to economic conditions rather than a fixed specie rule.

The break became concrete in "The Economic Consequences of Mr. Churchill" [Keynes, 1931], originally published in 1925. Keynes attacked Churchill's decision to return Britain to gold at the pre-war parity of $4.86, which overvalued sterling by roughly 10 percent and required British wages and prices to fall by that margin. The con-

sequences followed: industrial contraction, the coal dispute that became the General Strike of 1926, prolonged unemployment in the export industries. The episode confirmed that the price of monetary discipline under gold could be measured in unemployment rates and broken industries.

Keynes's reversal mirrors Friedman's in structure while inverting its conclusion. Both updated the same trade-off in response to the same cluster of events—the collapse of the pre-war monetary order, the Great War, the interwar deflation—but each concluded that the other's preferred failure mode was the greater danger. Keynes, observing from deflationary Britain, concluded that adjustment under specie discipline was the dominant risk. Friedman, composing the *Monetary History* with Schwartz [Friedman and Schwartz, 1963], concluded that discretionary monetary contraction was the dominant risk. Each was correct about the failure mode he emphasized. Eichengreen [1992] shows how the gold standard constrained monetary responses to the Depression; Friedman and Schwartz [1963] demonstrates how the Federal Reserve's discretionary failures deepened it. The divergent conclusions are the analytical content of the no-dominant-option result of Sargent [2011].

8.2 Marriner Eccles and the Treasury–Fed Accord of 1951

The structural parallel to Chase's arc belongs to Marriner Eccles. As Chairman of the Federal Reserve from 1934 to 1948, Eccles championed the wartime arrangement under which the Fed pegged Treasury bond prices—holding 90-day bill yields at $\frac{3}{8}$ percent and long bond yields at $2\frac{1}{2}$ percent. This was a deliberate crossing to fiscal dominance: the monetary authority became an unlimited buyer of government securities to keep borrowing costs low. Eccles understood the inflationary risk and accepted it as the price of war finance, saying so in congressional testimony and in his memoir [Eccles, 1951].

As wartime controls were lifted and inflation accelerated, the peg required the Fed to expand its balance sheet to prevent bond yields from rising—accommodating inflationary pressure rather than resisting it. By 1948, Eccles's arguments for restoring Federal Reserve independence had become a political liability. Truman demoted him from the chairmanship to an ordinary Board seat. Like Chase, who used his Chief Justiceship to attack the Acts he had designed, Eccles used his reduced position to press for monetary independence—and was penalized for it. The man who had designed the

fiscal-dominance arrangement was removed from its administration precisely because he had concluded it needed to end.

The Treasury–Fed Accord of March 4, 1951, negotiated by McCabe's successor William Martin, formally ended the peg [Hetzel and Leach, 2001]. The Fed regained authority to set interest rates without regard to Treasury borrowing costs. The Accord became the founding charter of American central bank independence [Meltzer, 2003].

Where Chase's *Hepburn* decision was reversed by *Knox* and extinguished by *Juilliard*, the Accord held. Several features appear decisive. First, the inflationary consequences of the peg were visible and politically salient: it hurt middle-class savers, so the coalition for monetary independence extended beyond financial elites. Second, the Korean War generated some pressure for Treasury accommodation but not the existential emergency of 1861–65. Third, the Fed had an immediately exercisable alternative—raise interest rates without any legislative action—whereas Chase needed the Court to strike down a statute. The Accord held because the inflation tax imposed by the peg had created a constituency for its removal that the Civil War greenbacks had not.

8.3 Richard Nixon and the End of Bretton Woods

Nixon built his political identity as the anti-Keynesian: his 1968 campaign attacked Johnson's "guns and butter" policy—Vietnam War spending combined with Great Society expansion without compensating tax increases. He opposed discretionary deficit spending and price controls. His initial Council of Economic Advisers chairman, Paul McCracken, was a monetarist-inflected fiscal conservative.

Bretton Woods—linking the dollar to gold at $35 per ounce and other currencies to the dollar—was the twentieth century's international analog of the gold standard: monetary dominance at the systemic level, with the dollar's gold peg constraining American fiscal and monetary policy. By 1971, persistent balance-of-payments deficits generated by Vietnam and the Great Society had depleted the gold stock and filled foreign central banks with dollars they were inclined to convert. The fiscal dominance latent in Johnson's spending had gradually undermined the monetary constraint, as Sargent–Wallace arithmetic would predict.

On August 15, 1971, Nixon announced the New Economic Policy: he simultane-

ously suspended dollar–gold convertibility, imposed a ninety-day freeze on wages and prices, and announced a Keynesian stimulus package designed to ensure re-election in 1972 [Matusow, 1998]. He told his economic advisers, "I am now a Keynesian in economics."

Nixon's reversal was not compelled by military emergency. The balance-of-payments pressure that broke Bretton Woods was fiscal: the accumulated legacy of "guns and butter" spending had inflated the dollar monetary supply relative to gold reserves. This is the peacetime version of the mechanism Sargent [2011] identifies: when fiscal dominance is the revealed preference of the political system, no external monetary anchor holds indefinitely, even without armed conflict.

Nixon announced the gold window closure as provisional. It became permanent. Gowa [1983] documents the political economy: the domestic fiscal commitments that caused the balance-of-payments crisis were not retracted, and without their retraction the gold price could not have been defended. The emergency of August 1971 became the permanent monetary constitution of the world it transformed.

9 Common Themes

Several themes recur across these reversals.

War and systemic crisis as forcing functions—and the clustering of reversals. The reversals documented here cluster at moments of acute fiscal and financial stress. The War of 1812 converted Clay and Madison simultaneously, and vindicated Gallatin's earlier administrative conclusions. The Civil War broke Chase, constructed the bond-backed currency system, and established the fiscal-dominance precedent that *Juilliard* would later entrench as permanent. Britain's participation in the Great War destroyed the international gold standard that Keynes had analyzed affirmatively in 1913 and set in motion the deflationary chain that drove his 1923 reversal. World War II created the wartime peg that placed Eccles at the fiscal-dominance extreme; postwar inflation drove him to the opposite pole. Nixon's 1971 reversal was the delayed fiscal consequence of Vietnam spending accumulated over Eisenhower, Kennedy, and Johnson administrations.

This clustering is not coincidental. In a steady fiscal environment, the costs of the

existing institutional arrangement are broadly distributed across time and may be tolerable. Under the concentrated demands of war or systemic panic, they become immediate and undeniable. Fiscal dominance—using monetary instruments to finance emergency expenditures—becomes irresistible when the alternative is defeat or systemic collapse. That is not an argument against institutional constraints; it is a reminder that any constraint must be evaluated by its behavior under the extreme stress cases that history periodically generates. *The dynamic dimension.* Sargent and Wallace [1981] and Sargent [2011] focus on steady-state comparisons of alternative institutional regimes. The historical record adds a dynamic dimension: transitions between regimes—forced by crisis—may matter as much as the steady states themselves. A monetary-dominant regime that collapses under wartime pressure may leave the fiscal authority worse off than a more flexible fiscal-dominant regime that never pretended to prohibit seigniorage. The value of an institutional commitment depends not only on its steady-state properties but on its resilience when tested.

The reflexivity of experience. Each reversal appealed to experience that the original position could not have anticipated. Clay and Madison could not have known in 1791 or 1811 what the War of 1812 would reveal about the fiscal cost of the absence of a national monetary institution. Friedman could not have known in 1948 what his and Schwartz's subsequent research would show about the monetary origins of the Depression, or what postwar Keynesian political economy would reveal about the systematic bias toward fiscal expansion. Keynes could not have known in 1913 what the Great War's fiscal demands would do to the monetary system he had described. Experience updated the effective parameters of the trade-off: the relative magnitude of fiscal and monetary risks, and the capacity of political institutions to manage them.

The constitutional versus the pragmatic argument. Chase's career and Madison's liquidation doctrine reveal a tension between constitutional arguments about where lines may be drawn and pragmatic arguments about where they should be drawn. The constitutional argument tries to fix the line against the pressures of the moment; the pragmatic argument allows the line to respond to circumstances. Madison resolved the tension by treating constitutional meaning as partly determined by experience. Chase resolved it by insisting the Constitution fixed the line absolutely—and discovered that the political economy would not sustain that absolutism. Peel's Bank Charter Act illustrates the same tension in statute: a rule rigid enough to

command credibility was not flexible enough to survive a genuine crisis, and Bagehot's response—institutionalizing suspension as lender-of-last-resort doctrine—was the most coherent resolution available.

The ratchet problem. Friedman's fixed money-growth rule was motivated partly by his assessment that coupling monetary and fiscal policy in a democracy reliably moves one way: toward more money creation, more inflation, more fiscal expansion. The record supports this. Clay and Madison, having supported the Second Bank, did not seek to use it for fiscal dominance; but Jackson's destruction of the Bank was followed by the pet-bank era, the suspension of 1837, and eventually the Legal Tender Acts. Nixon announced the gold window closure as temporary; it became permanent. Chase's emergency greenbacks were validated by *Juilliard* as a permanent attribute of sovereignty. The ratchet runs toward fiscal dominance when monetary institutions are weak; only durable, credible monetary commitments can reverse it—and such commitments are, as Eccles and Volcker both discovered, fragile under sustained fiscal pressure.

The impossibility of personal commitment. Chase's career shows that individuals cannot maintain institutional lines against political and fiscal pressure through the force of their convictions alone. Chase stated his monetary principles publicly for decades. Under sufficient pressure—the fiscal demands of a war for national survival—he abandoned them. Having abandoned them, he could not restore them from the bench, because the political forces that had driven the abandonment had not changed. Eccles faced the same discovery. Institutional lines must be maintained by institutions, not individuals; and as Madison's liquidation doctrine suggests, institutions themselves are mutable.

The asymmetry of conclusions from the same evidence. The same historical events drove Keynes and Friedman to opposite conclusions. The interwar monetary disorder drove Keynes toward managed money and fiscal flexibility, and drove Friedman toward a binding monetary rule. Both conclusions were correct about the failure mode each emphasized. This is the analytical content of the no-dominant-option result of Sargent [2011]: the correct institutional conclusion from the same historical experience depends on a judgment about which of two genuine risks is the greater—and that judgment can differ between equally careful observers reasoning from identical evidence.

10 Conclusion

The cases examined here do not resolve the trade-off between price-level stability and Ramsey-optimal fiscal efficiency identified in Sargent [2011]. They confirm that the trade-off is genuine, its terms shift with circumstances, and the optimal location of the institutional line has no permanent answer.

Two structural patterns emerge. First, reversals cluster at moments of systemic stress—wars, panics, depressions—when the costs of existing arrangements become undeniable and pressure toward fiscal dominance becomes irresistible. Second, reversals are rarely provisional: the emergency measure typically becomes the permanent settlement. The Legal Tender Acts were validated as a standing attribute of sovereignty in *Juilliard*. The gold window, announced as a temporary closure in 1971, was never reopened. Each new institutional line is drawn in response to the failures of its predecessor and generates its own successor crisis. The ratchet runs toward fiscal dominance, and durable monetary commitments are—as Eccles and Volcker both learned—fragile under sustained fiscal pressure.

The deeper lesson is not that institutions are useless but that no arrangement is secure. The question of where to draw the lines is structurally resistant to final resolution: the no-dominant-option result of Sargent [2011] is not a feature of particular parameter values but a consequence of the underlying trade-off itself. The reversals documented here—spanning two centuries, two countries, and actors as carefully reasoned as Madison, Chase, Keynes, and Friedman—are the predictable consequence of that structure. Anyone who believes the problem has been permanently solved has not been paying attention.

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